NIOSH Science Blog

Non-occupational Uses of Respiratory Protection – What Public Health Organizations and Users Need to Know

January 4, 2018 by Ronald Shaffer, PhD; Jaclyn Krah Cichowicz, MA; Ginger Chew, ScD; and LCDR Joy Hsu, MD, MS

The National Institute for Occupational Safety and Health (NIOSH) mission is to promote safe and healthful working conditions by conducting research and providing recommendations to diminish hazardous situations within occupational settings. However, there are situations in which the same respiratory hazards that can be faced by workers are also a potential concern to the public.

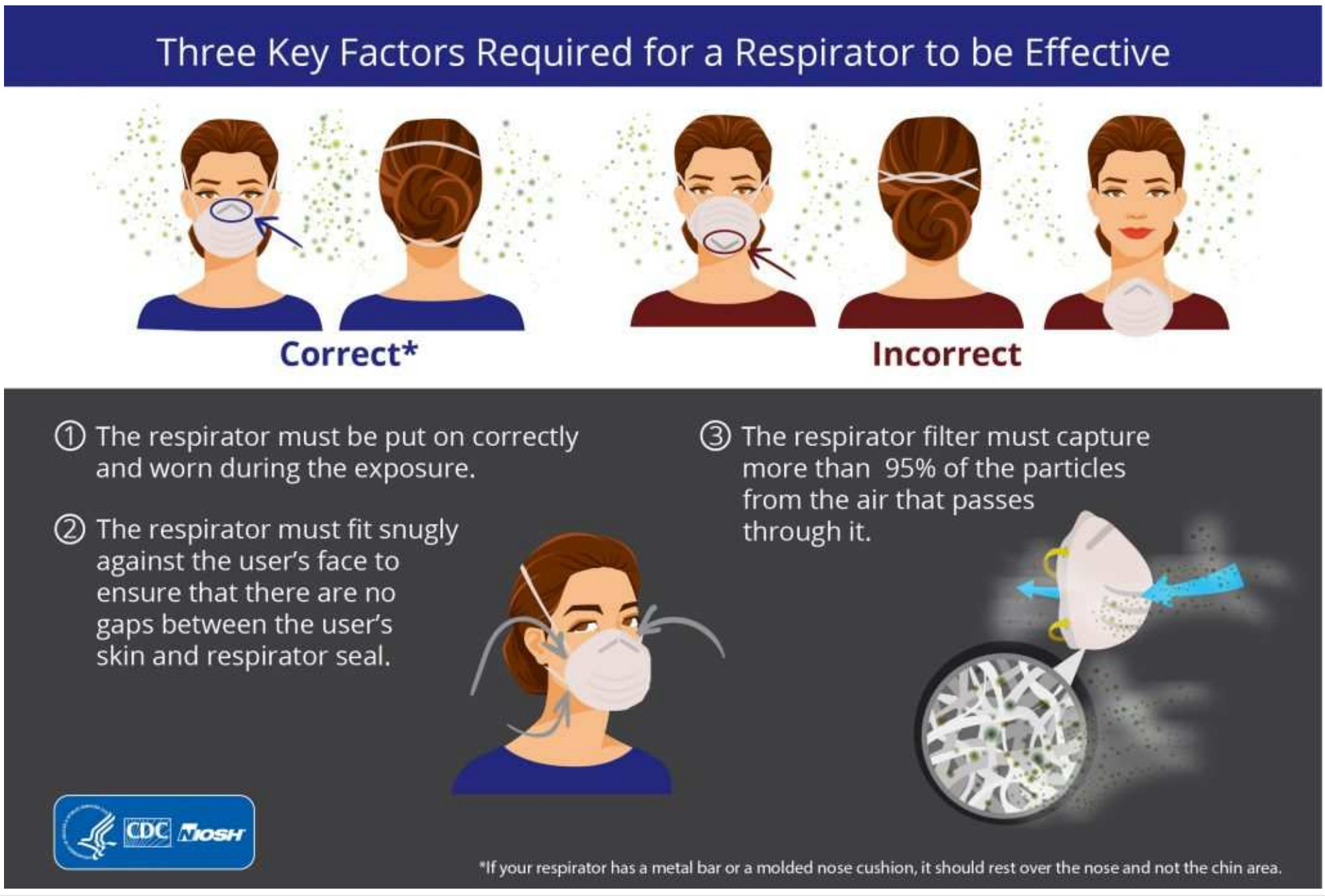
Examples of respiratory hazards of interest to the general public:

- Air pollution[1]
- Wildfires[2]
- Indoor mold growth after a flood[3]
- Bacteria and viruses[4]

A respirator is one method for reducing personal exposure to particulate respiratory hazards. NIOSH has been certifying respirators and conducting research on these devices for over 40 years. Along with our collaborators in government, industry, and academia, we have accumulated a wealth of knowledge on how respirators perform in workplace settings. In the last decade, respirator use by the general public has become a more frequent topic of debate as public health officials at the local, state, and federal level consider which public health and non-pharmaceutical interventions to recommend. In this blog, we have teamed with colleagues from across the Centers for Disease Control and Prevention (CDC) to assist state and federal public health agencies, and other safety professionals, by translating the lessons we've learned from respirator use at work to respirator use by the general public.

How Respirators Work

To appreciate the differences between workplace respirator use and personal respirator use, it is important to understand how these devices work. Three key factors are required for a respirator to be effective: (1) the respirator has to be put on correctly and worn during the exposure; (2) the respirator must fit the user's face snugly to minimize the number of particles that bypass the filter and get into the breathing zone through gaps between the user's skin and the respirator seal; and (3) the respirator filter needs to be highly effective at capturing particles that pass through the filter.



Respirator Use at Work

The Occupational Safety and Health Administration (OSHA) has regulations (29 CFR 1910.134) in place to ensure that all three of these items are addressed in occupational settings where respirator use is required. All employers with employees required to wear a respirator must (1) provide training which covers putting on a respirator and when the respirator should be used; (2) ensure that the respirator is both comfortable and fits the wearer using a fit test protocol conducted by a safety professional; and (3) use a NIOSH-approved respirator demonstrated to meet a certain level of filtration efficiency (e.g., 95% or more). In addition, workers in a complete respiratory protection program, receive a medical evaluation to ensure that they can safely wear their assigned respirator for the hazard and work task.

Numerous workplace protection factor studies have demonstrated that when used correctly, within the context of a complete respiratory protection program following OSHA 1910.134, any NIOSH-approved particulate respirator, regardless of make, model, or size, will provide a minimum level of protection.[5] For example, a N95 filtering facepiece respirator (FFR), when used properly, will reduce the amount of particles being inhaled at work to at least 1/10 of that in the room.

While FFRs have proven to be an essential component of occupational safety and health, their use is not without challenges. For example, most workers will find that wearing an FFR is not comfortable because a properly worn respirator will trap warm, moist air around your face. However, the effort required to move air in and out of the device is not noticeable for most people. Studies have shown that FFR use causes no additional physiological stress to most wearers, including pregnant women and does not contribute to heat stress .[6-8]

Voluntary Respirator Use at Work

OSHA's 1910.134 respiratory protection standard does allow for workplace situations in which respirator use is voluntary because the exposure is below an exposure limit. For example, some employees who have seasonal allergies may request a respirator for comfort when working outdoors, or an employee may request a respirator for use during non-hazardous exposures such as sweeping a dusty floor. To encourage respirator usage in these types of situations, OSHA only requires employers to ensure that the respirator does not present a health hazard to the worker. For FFRs, this means that training, fit testing, and medical evaluation are not required.

Respirator Use by the General Public

Respirator use by the general public is not subject to the same regulations required of employers in workplaces. Similar to voluntary use in workplaces, wearers will not have the benefits of formal training, fit testing, or medical evaluation. Additionally, while NIOSH-approved respirators are the standard within all relevant workplaces, for the general consumer there are many different types of respirators and facemasks available.[9] Because of these considerations, a few differences emerge consistent with the three key factors to effective respirator use:

- **Without training, respirators are likely to be worn incorrectly.** NIOSH conducted research in New Orleans after Hurricane Katrina in order to evaluate the correctness of N95 FFR donning by the public for mold remediation and found that only 24% of participants demonstrated proper donning technique.[10] Common mistakes included the clip (the metal band at the top of the FFR) not being pressed or tightened against the contours of the user's face, straps incorrectly placed, and putting the respirator on upside down. Users should follow manufacturers' guidance and the best practices below for guidance.
- **Without fit testing, it is not possible to know how well the respirator you use fits you.** Fit testing is a critical step as it increases the chances that you will receive expected levels of protection. Studies have shown that face seal leakage from a poor fit is the largest contributor to poor protection[11], aside from not wearing your respirator at all or wearing it incorrectly. See best practices below for fit testing recommendations.
- **Improvised devices such as bandanas and t-shirts, non-NIOSH-approved single-strap dust masks, and loose-fitting facemasks will not provide the same level of protection as a NIOSH-certified respirator.** The reason for this is two-fold.
    1. NIOSH-approved respirators provide higher levels of filter efficiency than improvised devices, facemasks, and dust masks.[12-14] This is not surprising as only the NIOSH-approved respirators are tested against a near "worst-case" aerosol challenge (i.e., mass median aerodynamic diameter particle of about 0.3 microns) and demonstrated to be at least 95% efficient. This ensures that filters in NIOSH certified respirators will very efficiently collect aerosols of all sizes and shapes, including air pollutants (such as PM2.5 which is particulate matter 2.5 microns or less) and biological aerosols. Other test methods exist for testing loose-fitting facemasks used in hospitals, but are far less stringent.[15] (More about worst-case testing in this NIOSH video, A Particle is a Particle.)
    2. Even without fit testing, studies have shown that users from the general public have a much greater chance of getting a good fit with a respirator (either a NIOSH-approved respirator or one that has also been cleared by the Food and Drug Administration (FDA) for use by the general public) than an improvised device, dust mask, or loose-fitting facemask.[16-21] This is because NIOSH-approved and/or FDA-cleared respirators are designed to be tight-fitting and meet minimum standards of construction. For example, one study[20] of eighteen different N95 FFRs found that without fit-testing 74% of the times the respirator was put on resulted in adequate levels of protection. These findings were highly dependent upon respirator model, with results ranging from 31% for the poorest-fitting models to 99% with the best fitting models. However, members of the general public wearing loose-fitting facemasks and improvised devices rarely approach the level of protection found by respirator wearers. For example, one study[21] found that on average only 1% of the times that a surgical mask was put on (range: 0% to 7% across 6 models) resulted in adequate levels of protection. This is not surprising as loose-fitting facemasks and improvised devices are not designed to seal tightly to the face and thus cannot prevent particles in the air from bypassing the filter and being drawn into the respiratory tract during normal breathing.

Best Practices

When N95 FFR use is appropriate, here are some best practices for maximizing the chances of getting good protection at home or during voluntary respirator use at work.

- Make sure your disposable N95 FFR has been approved by NIOSH or cleared by the FDA for public use.[22] You can determine if your respirator is NIOSH-approved by using our Searchable Certified Equipment List and by making sure that your respirator has the appropriate markings printed on the filter.
- Order from a reputable vendor; counterfeit devices are being sold via the internet.
- If you have been fit tested and use a respirator at work, then use the same respirator model and size at home. Studies have suggested that the more familiar you are with a respirator, the more likely you are to get a good fit.[23]

- Closely follow the manufacturer's instructions so that you put the respirator on correctly each time. A nice pictorial guide to respirator donning can be found here. The manufacturer's instructions on the respirator package or insert will help you to choose the best size of respirator and to wear it right. Clean-shaven faces will allow the respirator to fit tightly.
- Make sure the respirator fits properly, even if not fit tested, by performing a user seal check (Check out the OSHA/NIOSH video on donning/doffing and how to conduct a proper user seal check.)
- In general, no medical conditions automatically disqualify you from wearing an N95 FFR. However, if you are planning to wear a reusable half-facepiece or full-face respirator and have lung or heart disease, consider talking to your doctor or other medical provider first.
- If possible, take frequent breaks (every hour or so) away from the respiratory hazard. Use this time to stay hydrated and take the respirator off. This may be helpful for users that may not be used to wearing a respirator for long periods of time.
- Try to avoid touching the contaminated surface of the N95 FFR (the outer surface) so that you do not transfer potentially harmful particles to your hands.
- When done using the N95 FFR, dispose of it properly so that others (for example, children) cannot come in contact with it.
- Follow the manufacturer's recommendations for disposing of the respirator. N95 FFRs are disposable and designed to be worn only once. Some situations may allow you to "reuse" or put the same respirator on multiple times, but the respirator performance will degrade over time.

Summary

Thus, while respirator use outside of the workplace setting without training and fit testing is unlikely to provide the same level of protection for everyone as when all aspects of a respiratory program are followed, it does provide some health benefits to most healthy users[24-25] and can be considered a potential option by public health officials for reducing personal exposure to particulate respiratory hazards.

NOTE: The respirators we describe in this blog only filter-out particles, i.e., dusts and solid aerosols. If the environment includes toxic gases or vapors, a different type of respirator is needed that has a cartridge designed to filter the specific gas or vapor in the atmosphere; and if the environment is oxygen-deficient, a supplied-air type of respirator is needed. For more information about selecting the right respirator for the hazard please refer to the NIOSH respirator selection logic.

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See Also:

Breathe Easy, Part 1: How to Pick the Right Respirator for Your Emergency Kit

Breathe Easy, Part 2: How to Properly Use a Respirator in an Emergency

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The Centers for Disease Control and Prevention is addressing questions related to the Coronavirus Disease 2019 through CDC-INFO and on their webpage. As such, this blog has been closed to comments. Please visit https://www.cdc.gov/coronavirus/2019-ncov/index.html. You can find the most up-to-date information on the outbreak and get the latest answers to frequently asked questions. If you have specific inquiries, please contact CDC-INFO at https://wwwn.cdc.gov/dcs/contactus/form or by calling 800-232-4636. **If you have questions about PPE that are not related to Coronavirus Disease 2019, please contact us at PPEConcerns@cdc.gov.**

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Infectious Disease Resources, Respirators

7 comments on "Non-occupational Uses of Respiratory Protection – What Public Health Organizations and Users Need to Know"

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Excellent comprehensive information about the use of respirators especially concerning Donning and Doffing technique.
I will share with my team members.
Thanks.

It might be useful to put this document in an abbreviated form for workers/volunteers and translate it into other languages.

Thank you for pulling all of this information together. It is a great resource & since we know that the public is using respirators it is worthwhile to have accurate information available when questions arise.

Sometimes you need to go beyond the N95.
I was recently in a valley filled with wildfire smoke for a few days, and found the basic N95 mask wasn't much help. Local residents had been through wildfire smoke before, so appeared in a menagerie of respirators, many brightly colored. A local hardware store was almost sold out of their entire aisle of masks and respirators, but I managed to snag an N95 respirator, which helped more than the N95 mask, but not enough.
After this experience I'm very interested in respiratory protection above and beyond N95.

Excellent comment from Dr. Zafar "Excellent comprehensive information about the use of respirators"; nonetheless there's still more work to do on the subject as we uncertain how the respirators can beneficial to the working force.

We are facing questions on providing guidance as to should workers or the general population wear N95 during episodes of high Pm2.5 or if the AQI is high?

A realistic and a more representational of the population study should be conducted in order to understand the effectiveness of N95. It should also perhaps try to establish is what levels of AQI should the general population don N95.

What type of respirator's should first responder's be wearing? Should full face respirator's be considered?
If so, what would be the best model for protecting our first responder's from the CVID-19 virus. Trying to be proactive.

> Due to the nature of your inquiry and it's relation to Coronavirus Disease 2019 (COVID-19), please visit CDC's COVID-19 website. On the website, you can find the most up-to-date information on the outbreak, including resources for first responders, and get the latest answers to frequently asked questions. You can also contact CDC-INFO or by calling 800-232-4636.

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